The documentation and process conversion measures necessary to comply with this revision shall be completed by 4 January 1994

INCH POUND

MIL-S-19500/337F
4 October 1993
SUPERSEDING
MIL-S-19500/337E
8 June 1987

MILITARY SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING
TYPES 1N4153, 1N4153-1, 1N4153UR-1, AND 1N4534
JAN, JANTX, JANTXV, AND JANS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the detail requirements for silicon, switching diodes. Four levels of product assurance are provided for each device type as specified in MIL-S-19500 (see 6.3.2).

1.2 Physical dimensions. See figure 1 (similar to DO-35 and DO-34) and figure 2 (DO-213AA).

1.3 Maximum ratings.

| $V_{(BR)}$ | $V_{RWM}$ 1/ | $I_O$ 2/ | $I_{FSM}$ $t_p$ = 1 s | $I_{FSM}$ $t_p$ = 1 µs | $T_{OP}$ and $T_{STG}$ | $Z_{\theta JX}$ | $R_{\theta JL}$ | $R_{\theta JEC}$ |
|---|---|---|---|---|---|---|---|---|
| V dc | V (pk) | mA dc | A (pk) | A (pk) | °C | °C/W | °C/W | °C/W |
| 75 | 50 | 150 | 0.25 | 2.0 | -65 to +175 | 50 | 250 | 100 |

1/ This voltage may be exceeded for normal measuring intervals (< 1 minute) provided the current is limited to a maximum of 100 µA dc.
2/ Derate at 1.0 mA dc/°C above $T_A$ = 25°C.

1.4 Primary electrical characteristics at $T_A$ = +25°C, unless otherwise indicated.

| Limits | $V_{F1}$ $I_F$ = 100 µA dc | $V_{F2}$ $I_F$ = 250 µA dc | $V_{F3}$ $I_F$ = 1 mA dc | $V_{F4}$ $I_F$ = 2 mA dc | $V_{F5}$ $I_F$ = 10 mA dc | $V_{F6}$ $I_F$ = 20 mA dc |
|---|---|---|---|---|---|---|
| Min | 0.490 V dc | 0.530 V dc | 0.590 V dc | 0.620 V dc | 0.700 V dc | 0.740 V dc |
| Max | 0.550 V dc | 0.590 V dc | 0.670 V dc | 0.700 V dc | 0.810 V dc | 0.880 V dc |

| Limit | $I_{R1}$ $V_R$ = 50 V dc | $I_{R2}$ $V_R$ = 50 V dc $T_A$ = 150°C | C $V_R$ = 0 V dc f = 1 MHz | $t_{rr}$ $I_F$ = $I_R$ = 10 mA dc $R_L$ = 100 ohms, $I_{RR}$ = 1.0 mA dc |
|---|---|---|---|---|
| Min | --- | --- | --- | --- |
| Max | 50 nA dc | 50 µA dc | 2.0 pF | 4 ns |

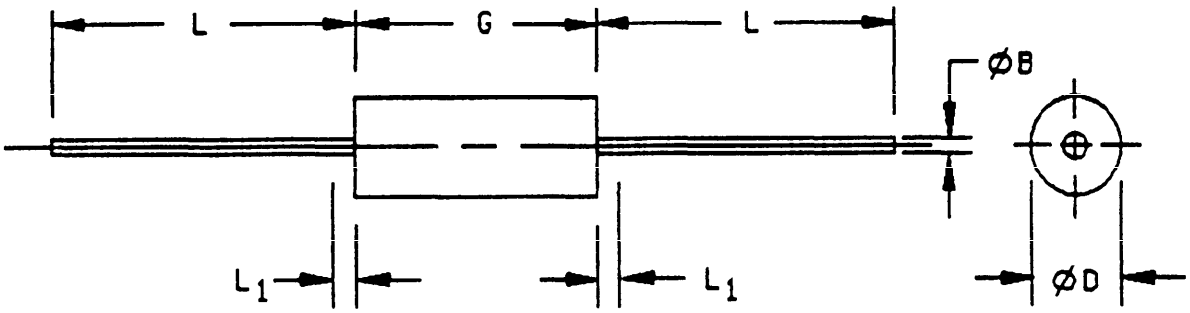
Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Commander, Defense Electronics Supply Center, ATTN: DESC-ECT, 1507 Wilmington Pike, Dayton, OH 45444-5270, by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A

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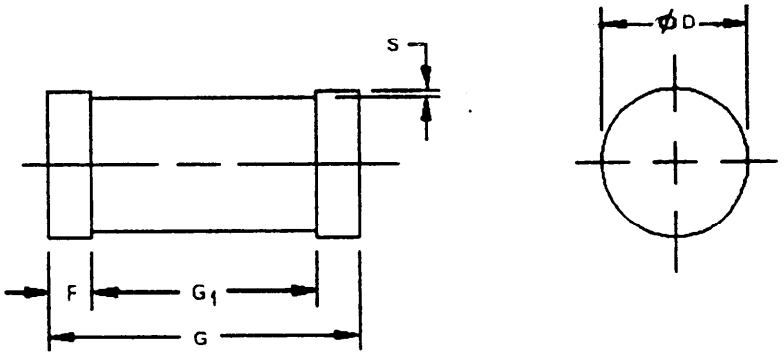

| Types | Dimensions | | | | | Notes |
|---|---|---|---|---|---|---|
| | Ltr | Inches | | Millimeters | | |
| | | Min | Max | Min | Max | |
| 1N4153<br>1N4153-1 | G | .140 | .180 | 3.56 | 4.57 | 3 |
| | ϕD | .056 | .075 | 1.42 | 1.91 | 3 |
| | L | 1.000 | 1.500 | 25.40 | 38.10 | |
| | L1 | | .050 | | 1.27 | 4 |
| | ϕB | .018 | .022 | 0.46 | 0.56 | |
| 1N4534 | G | .080 | .120 | 2.03 | 3.05 | 3 |
| | ϕD | .050 | .075 | 1.27 | 1.91 | 3 |
| | L | 1.000 | 1.500 | 25.40 | 38.10 | |
| | L1 | | .050 | | 1.27 | 4 |
| | ϕB | .018 | .022 | 0.46 | 0.56 | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Package contour optional within ϕD and length G. Heat slugs if any, shall be included within this cylinder but shall not be subject to the minimum limit of ϕD.
4. Lead diameter not controlled in zones L1 to allow for flash, lead finish build-up and minor irregularities other than heat slugs.

FIGURE 1. Physical dimensions for 1N4153-1 (DO-35) and 1N4534 (DO-34).



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| Symbol | Dimensions | | | |
|---|---|---|---|---|
| | Inches | | Millimeters | |
| | Min | Max | Min | Max |
| ØD | .063 | .067 | 1.60 | 1.70 |
| F | .016 | .022 | 0.41 | 0.55 |
| G | .130 | .146 | 3.30 | 3.70 |
| $G_1$ | .100 ref | | 2.54 ref | |
| S | .001 min | | 0.03 min | |

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.

FIGURE 2. Physical dimensions for 1N4153UR-1 (DO-213AA).



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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATION

MILITARY

MIL-S-19500 - Semiconductor Devices, General Specification for.

STANDARD

MILITARY

MIL-STD-750 - Test Methods for Semiconductor Devices.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Document Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19120-5094.)

2.2 Order of precedence. In the event of a conflict between the text of this document and the references cited herein (except for related associated detail specifications, specification sheets, or MS standards), the text of this document shall take precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-S-19500, and as specified herein.

3.2 Abbreviations, symbols, and definitions. Abbreviations, symbols, and definitions used herein shall be as specified in MIL-S-19500.

3.3 Design, construction, and physical dimensions. The design, construction, and physical dimensions shall be as specified in MIL-S-19500, and figures 1 and 2 herein.

3.3.1 Lead finish. Lead finish shall be solderable in accordance with MIL-STD-750, MIL-S-19500. Where a choice of lead finish is desired, it shall be specified in the acquisition document.

3.3.2 1N4153-1 construction. This device shall be constructed utilizing high temperature metallurgically bonded double plug construction in accordance with the requirements of category I,II, or III (see MIL-S-19500).

3.3.3 JANS construction. JANS construction shall be metallurgically bonded, double plug, non-cavity thermally matched package. JANS metallurgical bonds are limited to category I only in accordance with MIL-S-19500.

3.4 Marking. Marking shall be in accordance with MIL-S-19500.

4. QUALITY ASSURANCE PROVISIONS

4.1 Sampling and inspection. Sampling and inspection shall be in accordance with MIL-S-19500, and as specified herein.



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4.2 Qualification inspection. Qualification inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.3 Screening (all levels). Screening shall be in accordance with MIL-S-19500 (table II), and as specified herein. The following measurements shall be made in accordance with table I herein. Devices that exceed the limits of table I herein shall not be acceptable.

| Screen (see table II of MIL-S-19500) | Measurement | | |
|---|---|---|---|
| | JANS | JANTX and JANTXV levels | JAN Level |
| 3a | Temperature cycling | Temperature cycling | Temperature cycling (per MIL-S-19500 JANTX level) |
| 3c 1/ | Thermal impedance (see 4.5.2) | Thermal impedance (see 4.5.2) | Thermal impedance (see 4.5.2) |
| 9 | $I_{R1}$ and $V_{F6}$ | Not applicable | Not applicable |
| 11 | $I_{R1}$ ; $V_{F6}$ ; $\Delta I_{R1}$ ≤ 100 percent of initial value or 25 nA dc, whichever is greater; $\Delta V_{F6}$ ≤ 25 mV dc change from initial value | $I_{R1}$ and $V_{F6}$ | |
| 12 | See 4.3.1 | See 4.3.1<br>t = 48 hours (min) | |
| 13 2/<br>3/ | Subgroups 2 and 3 of table I herein; $\Delta I_{R1}$ ≤ 100 percent of initial value or 25 nA dc, whichever is greater; $\Delta V_{F6}$ ≤ 25 mV dc change from initial value. | Subgroup 2 of table I herein; $\Delta I_{R1}$ ≤ 100 % of initial value or 25 nA dc, whichever is greater; $\Delta V_{F6}$ ≤ 25 mV dc change from initial value. | |

1/ Thermal impedance may be performed any time after temperature cycling is completed in screen 3.

2/ Thermal impedance does not need to be repeated in step 13 if previously done in screening.

3/ PDA = 5%.

4.3.1 Power burn-in conditions. Devices shall be operated under one of the following conditions:

$T_A$ = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5.

Condition 1: $I_O$ = 150 mA dc, $V_R$ = 50 V (pk).

Condition 2: $I_F$ = 175 mA dc.

4.4 Quality conformance inspection. Quality conformance inspection shall be in accordance with MIL-S-19500, and as specified herein.

4.4.1 Group A inspection. Group A inspection shall be conducted in accordance with MIL-S-19500, and table I herein. (End-point electrical measurements shall be in accordance with the applicable steps of table II herein.)



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4.4.2 Group B inspection. Group B inspection shall be conducted in accordance with the conditions specified for subgroup testing in table IVa (JANS) and table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

4.4.2.1 Group B inspection, table IVa (JANS) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 4066 | Not applicable. |
| B4 | 1037 | $I_O$ = 150 mA dc; $V_R$ = 50 V (pk); $T_A$ = 25°C; $t_{on}$ = $t_{off}$ = 3 minutes minimum for 2,000 cycles. |
| B5 | 1027 | $I_O$ = 150 mA dc; $V_R$ = 50 V (pk); $T_A$ = 125°C |
| B6 | 4081 | $R_{\theta JL}$ at L = .375, ≤ 250°C/W, $R_{\theta JEC}$ at L = 0, ≤ 100°C/W. |

4.4.2.2 Group B inspection, table IVb (JAN, JANTX, and JANTXV) of MIL-S-19500.

| Subgroup | Method | Condition |
|---|---|---|
| B3 | 1027 | $I_O$ = 150 mA dc, $V_R$ = 50 V (pk), $T_A$ = Room ambient as defined in the general requirements of MIL-STD-750, paragraph 4.5. |
| B5 | 4081 | $R_{\theta JL}$ at L = .375" ≤ 250°C/W, $R_{\theta JEC}$ at L = 0, ≤ 100°C/W. |

4.4.4 Group C inspection. Group C inspection shall be conducted in accordance with the conditions specified for subgroup testing in table V of MIL-S-19500, and as follows. Electrical measurements (end points) and delta requirements shall be in accordance with the applicable steps of table II herein.

| Subgroup | Method | Condition |
|---|---|---|
| C2 | 2036 | (Not applicable for UR suffix version devices.)<br>Test condition A; weight = 4 lbs.; t = 15 s.<br>Test condition E; weight = 16 oz. |
| C6 | 1026 | $I_O$ = 150 mA dc, $V_R$ = 50 V (pk). |

4.5 Methods of inspection. Methods of inspection shall be as specified in the appropriate tables and as follows:

4.5.1 Pulse measurements. Conditions of pulse measurements shall be specified in paragraph 4.3.2.1 of MIL-STD-750.

4.5.2 Thermal impedance $Z_{\theta JX}$ measurements for screening. The $Z_{\theta JX}$ measurements shall be performed in accordance with MIL-STD-750, method 3101. The maximum limit (not to exceed the group A, subgroup 2 limit) for $Z_{\theta JX}$ in screening (table II of MIL-S-19500) shall be derived by each vendor by means of statistical process control. When the process has exhibited control and capability, the capability data shall be used to establish the fixed screening limit. In addition to screening, once a fixed limit has been established, monitor all future sealing lots using a random five piece sample from each lot to be plotted on the applicable X, R chart. If a lot exhibits an out of control condition, the entire lot shall be removed from the line and held for Engineering evaluation and disposition.

| | | |
|---|---|---|
| a. | $I_M$ measurement current | 1 mA - 10 mA. |
| b. | $I_H$ forward heating current | 300 mA (min). |
| c. | $t_H$ heating time | 10 ms. |
| d. | $t_{MD}$ measurement delay time | 100 µs maximum. |



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TABLE I. Group A inspection.

| Inspection 1/ | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 1 | | | | | | |
| Visual and mechanical inspection | 2071 | | | | | |
| Subgroup 2 | | | | | | |
| Thermal impedance | 3101 | See 4.5.2 | $Z_{\theta JX}$ | | 50 | °C/W |
| Forward voltage | 4011 | $I_F$ = 100 µA dc | $V_{F1}$ | 0.49 | 0.55 | V dc |
| Forward voltage | 4011 | $I_F$ = 250 µA dc | $V_{F2}$ | 0.53 | 0.59 | V dc |
| Forward voltage | 4011 | $I_F$ = 1 mA dc | $V_{F3}$ | 0.59 | 0.67 | V dc |
| Forward voltage | 4011 | $I_F$ = 2 mA dc | $V_{F4}$ | 0.62 | 0.70 | V dc |
| Forward voltage | 4011 | $I_F$ = 10 mA dc | $V_{F5}$ | 0.70 | 0.81 | V dc |
| Forward voltage | 4011 | $I_F$ = 20 mA dc | $V_{F6}$ | 0.74 | 0.88 | V dc |
| Reverse current | 4016 | DC method; $V_R$ = 50 V dc | $I_{R1}$ | | 50 | nA dc |
| Breakdown voltage | 4021 | $I_R$ = 5 µA dc | $V_{BR1}$ | 75 | | V dc |
| Subgroup 3 | | | | | | |
| High temperature operation: | | $T_A$ = +150°C | | | | |
| Reverse current | 4016 | DC method<br>$V_R$ = 50 V dc | $I_{R2}$ | | 50 | µA dc |
| Low temperature operation: | | $T_A$ = -55°C | | | | |
| Breakdown voltage | 4021 | $I_R$ = 5 µA dc | $V_{(BR)2}$ | 75 | | V dc |
| Subgroup 4 | | | | | | |
| Junction capacitance | 4001 | $V_R$ = 0 V dc, f = 1 MHz<br>$V_{sig}$ = 50 mV (pk to pk), max | C | | 2.0 | pF |
| Reverse recovery time | 4031 | Condition A; C ≥ 1 nF<br>$I_F = I_R$ = 10 mA dc<br>$i_{R(REC)}$ = 1.0 mA<br>R ≥ 1000 Ω, $R_L$ = 100 Ω | $t_{rr}$ | | 4 | ns |
| Subgroup 5 | | | | | | |
| Not applicable | | | | | | |

See footnote at end of table.



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TABLE I. Group A inspection - Continued.

| Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|
| | Method | Conditions | | Min | Max | |
| Subgroup 6 | | | | | | |
| Surge current | 4066 | $I_F$ = 150 mA dc;<br>$T_A$ = 25°C<br>$I_{FSM}$ = 2 A(pk)<br>ten 1 μs surges,<br>1 surge/min. | | | | |
| Electrical measurements | | See table II, steps 1 and 3 | | | | |
| Subgroup 7 | | | | | | |
| Not applicable | | | | | | |

1/ For sampling plan, see MIL-S-19500.



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TABLE II. Groups A, B, and C electrical measurements.

| Step | Inspection | MIL-STD-750 | | Symbol | Limits | | Unit |
|---|---|---|---|---|---|---|---|
| | | Method | Conditions | | Min | Max | |
| 1. | Forward voltage | 4011 | $I_F$ = 20 mA dc | $V_{F6}$ | .74 | .88 | V dc |
| 2. | Forward voltage | 4011 | $I_F$ = 20 mA dc | $V_{F7}$ | .71 | .91 | V dc |
| 3. | Reverse current | 4016 | DC method; $V_R$ = 50 V dc | $I_{R1}$ | | 50 | nA dc |
| 4. | Reverse current | 4016 | DC method; $V_R$ = 50 V dc | $I_{R1}$ | | 100 | nA dc |
| 5. | Forward current | 4011 | $I_F$ = 20 mA dc | $\Delta V_{F6}$ | ≤ 25 mV dc change from initial value | | |
| 6. | Reverse current | 4016 | DC method, $V_R$ = 50 V dc | $\Delta I_{R1}$ | ≤ 100% of initial value or 25 nA dc, whichever is greater | | |

1/ The electrical measurements for table IVa (JANS) of MIL-S-19500 are as follows:

a. Subgroup 3, see table II herein, steps 1 and 3.

b. Subgroups 4 and 5, see table II herein, steps 1, 3, 5, and 6.

2/ The electrical measurements for table IVb (JAN, JANTX and JANTXV) of MIL-S-19500 are as follows:

a. Subgroup 2, see table II herein, steps 1 and 3.

b. Subgroups 3 and 6, see table II herein, steps 2 and 4.

3/ The electrical measurements for table V of MIL-S-19500 are as follows:

a. Subgroups 2 and 3, see table II herein, steps 1 and 3.

b. Subgroup 6, see table II herein, steps 1, 3, 5, and 6 (JANS); steps 2 and 4 (JAN, JANTX, and JANTXV).



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5. PACKAGING

5.1 Packaging requirements. The requirements for packaging shall be in accordance with MIL-S-19500.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Notes. The notes specified in MIL-S-19500 are applicable to this specification.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of the specification.

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1).

c. Lead finish (see 3.3.1).

d. Product assurance level and type designation.

6.3 Changes from previous issue. Asterisks are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - EC
Air Force - 17
NASA - NA

Review activities:
Army - AR, MI, SM
Navy - AS, CG
Air Force - 19, 85, 99
DLA - ES

Preparing activity:
Navy - EC

Agent:
DLA - ES

(Project 5961-1442)



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1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.

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3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

| I RECOMMEND A CHANGE: | 1. DOCUMENT NUMBER<br>MIL-S-19500/337F | 2. DOCUMENT DATE (YYMMDD)<br>93/10/04 |
|---|---|---|

3. DOCUMENT TITLE SEMICONDUCTOR DEVICE, DIODE, SILICON, SWITCHING TYPES 1N4153, 1N4153-1, 1N4153UR-1, AND 1N4534 JAN, JANTX, JANTXV, AND JANS

4. NATURE OF CHANGE (Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)

5. REASON FOR RECOMMENDATION

6. SUBMITTER

| a. NAME (Last, First, Middle Initial) | b. ORGANIZATION | |
|---|---|---|
| c. ADDRESS (Include Zip Code) | d. TELEPHONE (Include Area Code)<br>(1) Commercial<br>(2) AUTOVON<br>(If applicable) | 7. DATE SUBMITTED<br>(YYMMDD) |

8. PREPARING ACTIVITY

| a. NAME<br>Alan Barone | b. TELEPHONE (Include Area Code)<br>(1) Commercial<br>513-296-6048<br>(2) AUTOVON<br>986-6048 |
|---|---|
| c. ADDRESS (Include Zip Code)<br>DEFENSE ELECTRONICS SUPPLY CENTER<br>1507 Wilmington Pike<br>Dayton, OH 45444-5270 | IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT:<br>Defense Quality and Standardization Office<br>5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466<br>Telephone (703) 756-2340 AUTOVON 289-2340 |



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